

Healthcare Analysis Dashboard

- Power BI project converting raw ER records into an interactive dashboard for monitoring admissions, wait times, demographics, referrals, and satisfaction to support data-driven decisions.



Dataset Overview

Total Records

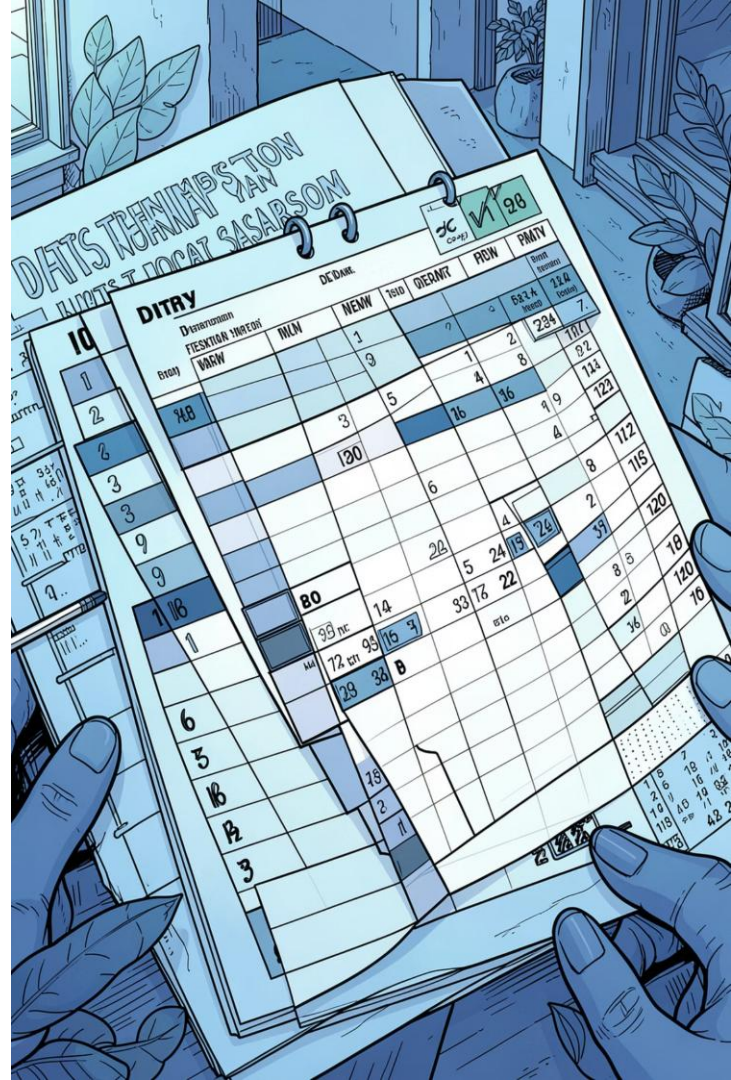
9,216 patient records across 12 original columns.

Derived Time Columns

Year, Month Number, Month Name, Day extracted from admission date.

Full Name

Created by combining Patient First Initial and Patient Last Name.



[illegible]

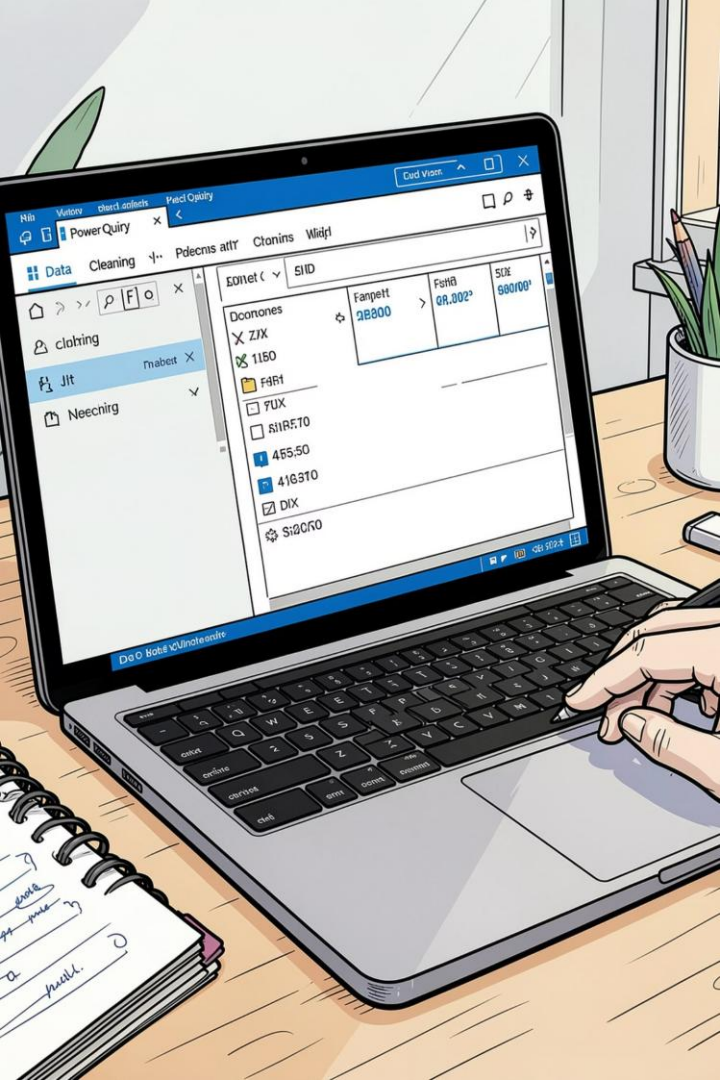
Patient ID, Full Name, Gender, Age, Race



Department Referral, Admission Status, Wait Time, Satisfaction Score



Admission Date, Monthly trends, Year-wise analysis



Data Preparation

- Imported Excel dataset into Power BI and validated data types.
- Created Full Name and extracted date components for time analysis.
- Handled blanks in Department Referral and Satisfaction fields.
- Removed formatting issues and prepared data for visuals.

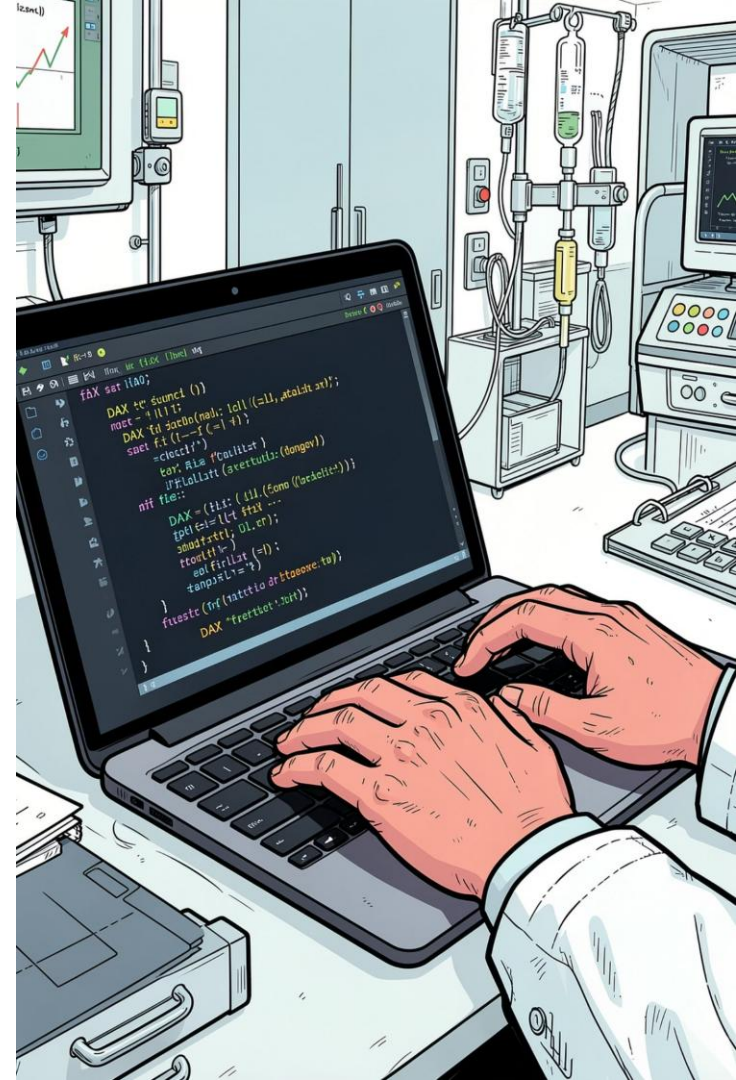
Calculated Columns & DAX

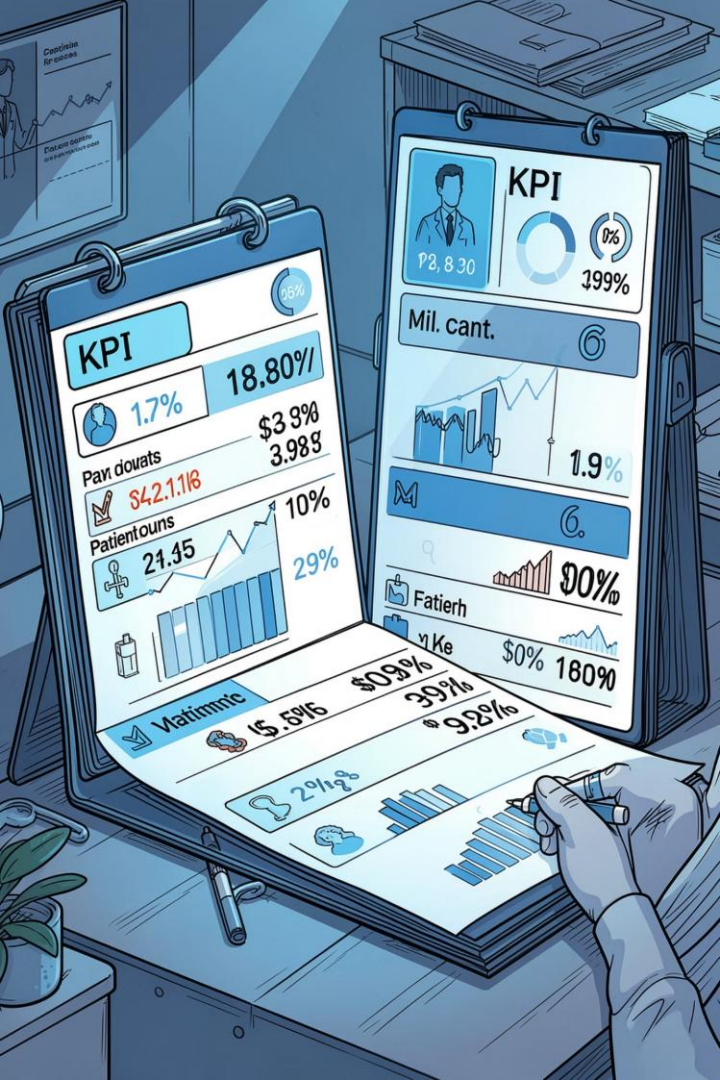
Full Name

Full Name = [Patient First Initial] & " " & [Patient Last Name]

Year

```
Year = YEAR([Patient  
Admission Date])
```





KPI Measures



Total Patients

COUNT('Hospital
ER_Data'[Patient ID])
— shows total ER
visits.



Average Waiting Time

AVERAGE('Hospital
ER_Data'[Patient Wait
Time]) — average
minutes waited.



Avg Satisfaction Score

AVERAGE('Hospital
ER_Data'[Patient
Satisfaction Score])
— overall experience.



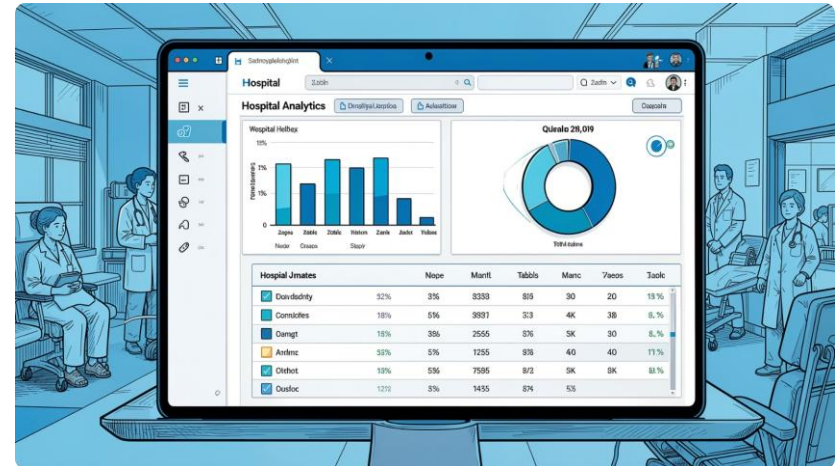
Total Admissions

CALCULATE(COUNT(...), 'Hospital ER_Data'[Admission Status] = "Admitted")

Dashboard Visuals

Charts included:

- Horizontal bar — Department referral counts
- Donut — Patient distribution by gender
- Column — Monthly record count
- Bar — Department-wise satisfaction
- Table — Detailed patient records



Key Insights

Patient Growth

Total patients rose 11.9% from March to April; April saw increased visits.

Highest Wait Time

Neurology average wait time: 55 minutes — suggests staffing or scheduling issues.

Top Satisfaction

Orthopedics satisfaction score: 4.6 / 5 — strongest patient experience.



Key Metrics Snapshot

9,216

Total Patients

35.3

Avg Wait Time (mins)

4.2

Avg Satisfaction (/5)

8

Total Departments

480

Case Management Records

Recommendations & Conclusion

- Increase staff in high-wait departments (Neurology) to reduce wait times.
- Allocate more resources to General Practice due to high referrals.
- Monitor satisfaction continuously and act on department-level feedback.
- Use monthly trends for staffing and resource planning.
- Leverage demographic patterns for personalized care delivery.



Conclusion: Power BI transformations, calculated columns, and DAX measures produced an interactive ER dashboard that supports operational and strategic decision-making.